ETH zürich

Mobile Computing

Emerging Technologies
Non-Terrestrial Satellite Networks

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Course Outline

Introduction

1. Data and Signals
2. Data Communication
3. Mobile Systems
4. Mobile Algorithms
5. Mobile AI
6. Cloud Computing

Tournament

Introduction to Satellite Networking

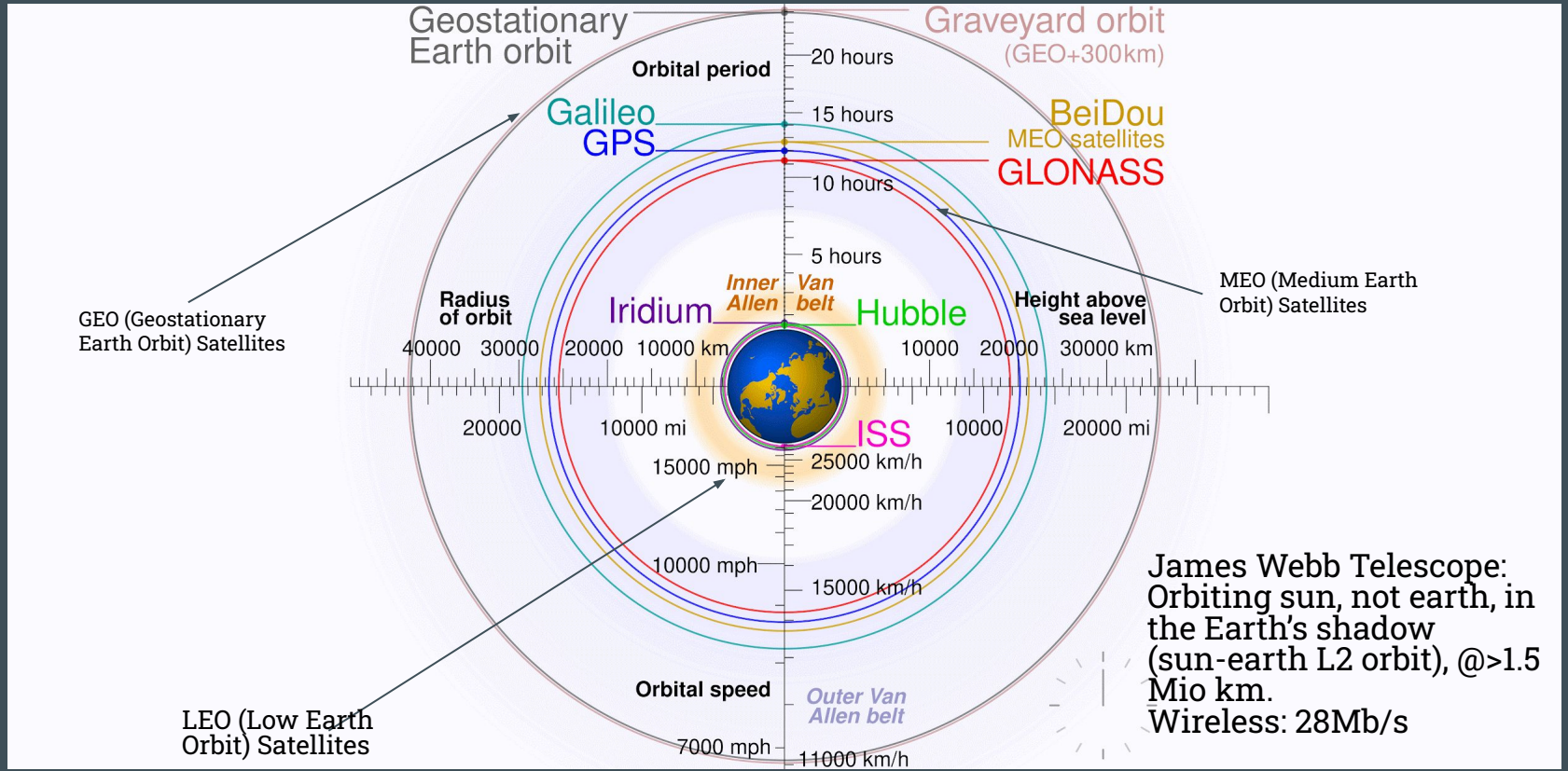


Image Source: https://commons.wikimedia.org/wiki/File:Comparison_satellite_navigation_orbits.svg

Low Earth Orbit

- 200–2,000 km above Earth
- Satellite Internet
  - Starlink, OneWeb, and similar for low-latency coverage
- Earth Observation
  - Imaging (e.g., weather satellites, climate monitoring, and remote sensing for agriculture or natural disaster mgmt.)
- Science and Research
  - Platforms like the International Space Station (ISS), Tiangong space station, Hubble

Medium Earth Orbit

- 2,000–35,786 km above Earth
- Global Navigation Systems
  - GPS, Galileo, and GLONASS satellites operate in MEO to provide precise positioning and navigation.
- Communication Satellites
  - Some systems (e.g., O3b Networks) use MEO to provide decent latency internet for underserved areas. This is becoming less relevant with the new LEO constellations.

Geostationary Earth Orbit

- 35,786 km above Earth
  - Telecommunications: TV broadcasting, satellite phones, and long-distance communication.
  - Weather Monitoring: Geostationary weather satellites like GOES continuously monitor large regions of Earth.
  - Military Communications: Used for secure and wide-area military networks.

Increasing number of payloads

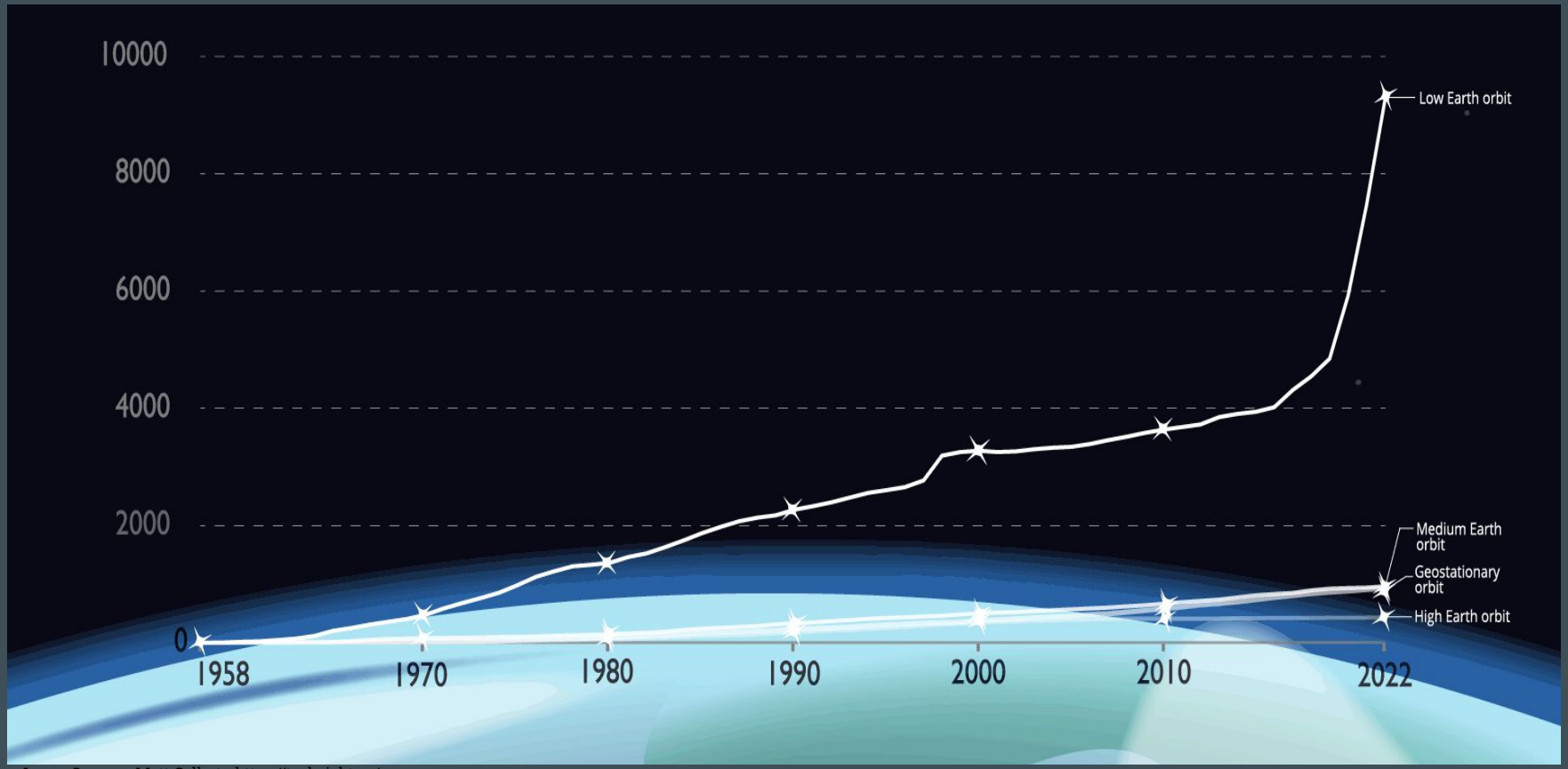


Image Sources: Matt Gellert - https://techsight.co/

Advantages of Satellite Networking

- Covers large areas, including difficult to reach locations
- Resilience to disasters compared to terrestrial infrastructure
- Speed of light ~33% faster vs fiber
  - NY-> London 22ms << 33ms

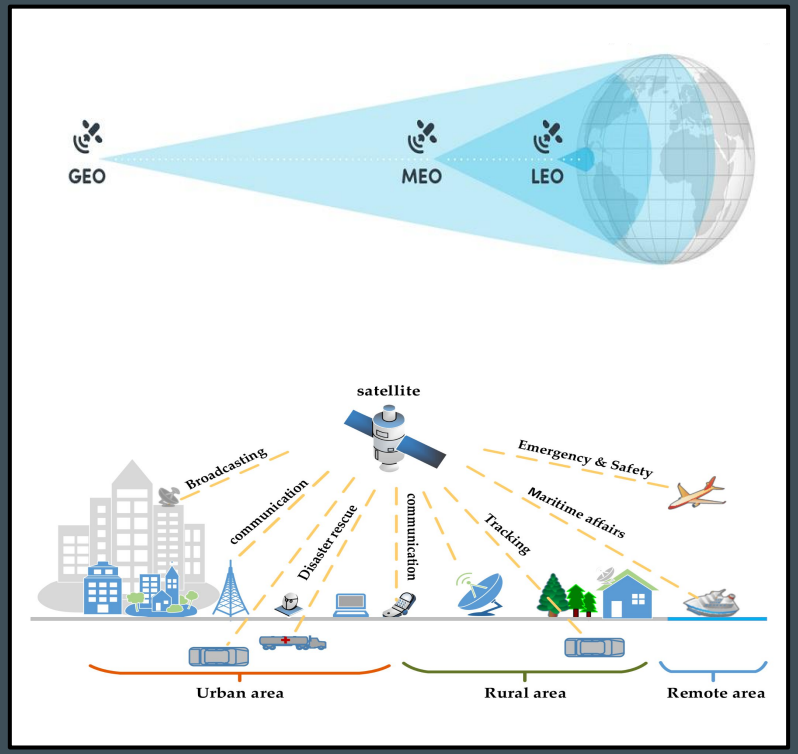


Image Sources: https://euroskypark.com/ and Modeling and Performance Analysis of Satellite Network Moving Target Defense System with Petri Nets

Disadvantages

- High latency (mainly for GEO)
- Cost of deployment, and maintenance
- Space debris
- Rain / atmospheric attenuation
- Atmospheric drag (LEO)

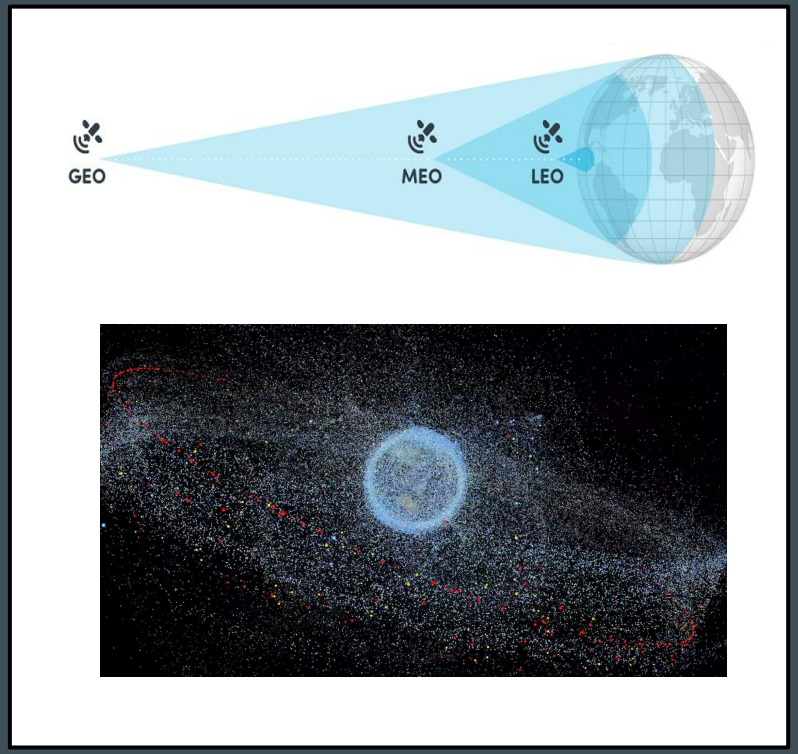


Image Sources: https://euroskypark.com/ and ESA (https://www.esa.int/ESA_Multimedia/Videos/2019/02/Distribution_of_space_debris_in_orbit_around_Earth

Satellite Networking Frequencies

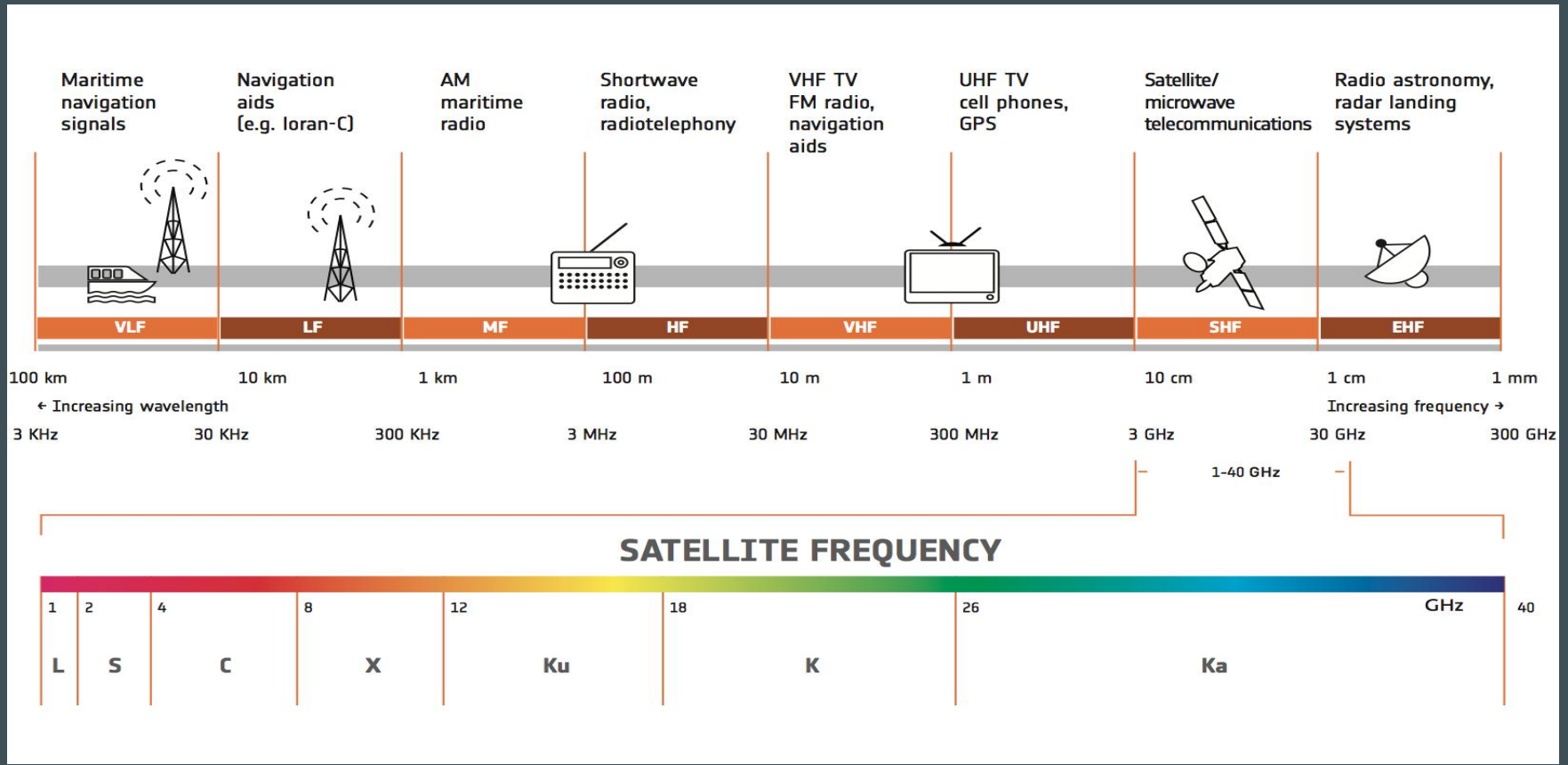
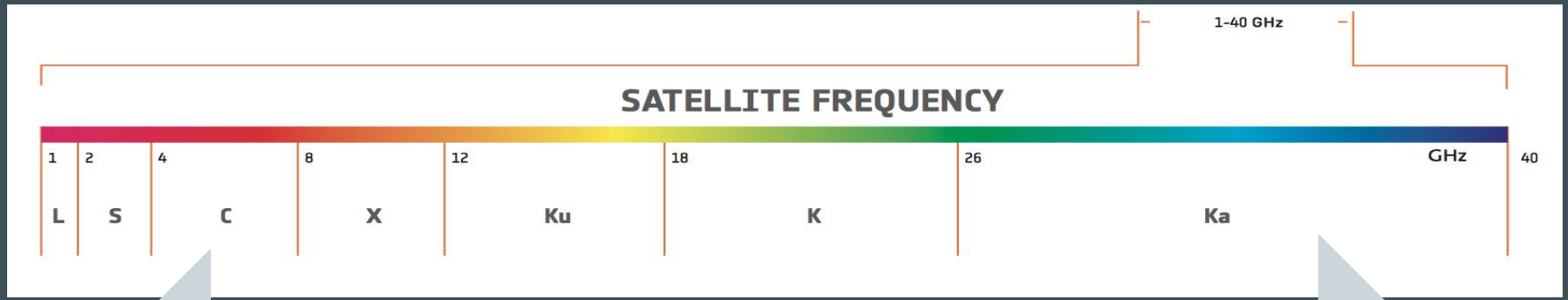


Image Sources: ESA - https://www.esa.int/Applications/Connectivity_and_Secure_Communications/Satellite_frequency_bands

Satellite Networking Frequencies



Better Propagation
Longer Range
More Reliable in Bad Weather

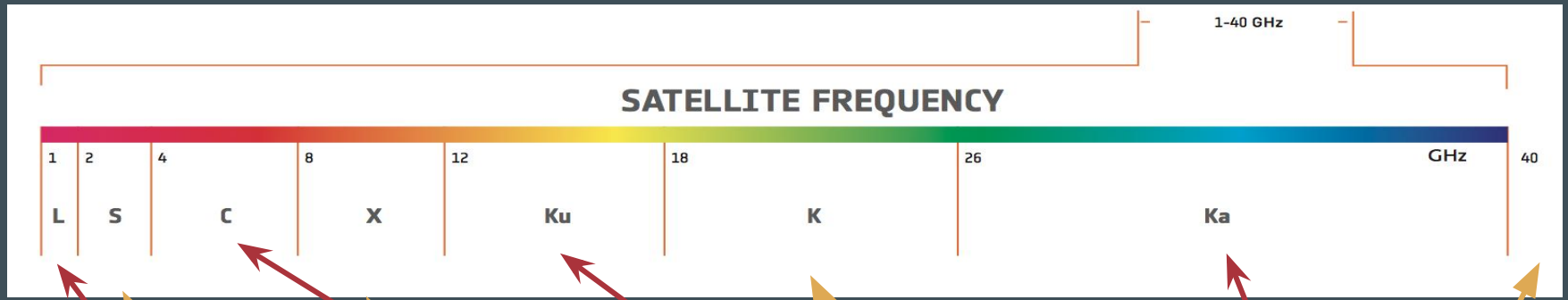
Higher Bandwidth
Smaller Antennas
Better for Point-to-Point
Communications

Lower Bandwidth
Larger Antennas

Higher Power Requirements
Susceptible to Rain Fade

Image Sources: ESA - https://www.esa.int/Applications/Connectivity_and_Secure_Communications/Satellite_frequency_bands

Satellite Networking Frequencies



GPS Systems

Weather Satellites

Satellite TV, long-distance telephone

Military Comms

Satellite Internet, Satellite TV

Not used much due to the 22 GHz water vapor absorption line

Satellite Internet (Starlink, Project Kuiper)

Laser Links are starting to be used (>200THz) in some scenarios

Image Sources: ESA - https://www.esa.int/Applications/Connectivity_and_Secure_Communications/Satellite_frequency_bands

Satellite Networking Frequencies

- International Telecommunication Union (ITU)
  - ITU's Radiocommunication Sector allocates frequency bands for space-based services and coordinates frequency assignments globally.
  - ITU also ensures that satellite operators comply with international agreements to avoid interference and ensure equitable access to spectrum for all countries.

Satellite Networking Frequencies

- National Regulatory Authorities (FCC in the US, Ofcom in the UK)
  - Each country has its own national regulatory authority responsible for managing and allocating spectrum within its borders, including the regulation of satellite frequencies.
  - These agencies ensure that satellite systems operating in their jurisdiction comply with national laws and ITU regulations, and they grant licenses to satellite operators.

Satellite Networking Frequencies

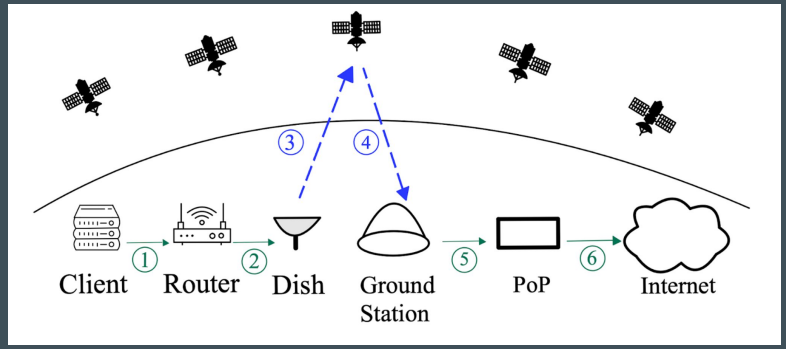
- Different satellites can have very different requirements and applications
  - GPS satellites need to send very little data in a broadcast-only manner but need to be resilient to atmospheric events
  - TV satellites have higher bandwidth requirements and need to have reliable, continuous coverage to a fixed area (one GEO satellite has roughly coverage to ⅓ of Earth at any given time).

Satellite Networking Frequencies

Satellite Type	Frequency Bands	Typical Bandwidth	Bandwidth (Mbps/Gbps)	Latency
GPS Satellites	L1, L2 (L-band: 1-2 GHz)	1-2 MHz per signal	**< 1 Mbps** (total system)	50-200 ms (signal travel)
TV Satellites (GEO)	**C-band** (3.7-4.2 GHz)	36 MHz to 72 MHz per transponder	**~ 36-72 Mbps** (per transponder)	500 ms (round trip)
	Ku-band (12-18 GHz)	36 MHz to 72 MHz per transponder	**~ 72 Mbps to 500 Mbps** (per transponder)	500 ms (round trip)
	Ka-band (26.5-40 GHz)	500 MHz to 1 GHz per transponder	**~ 500 Mbps to 1 Gbps** (per transponder)	500 ms (round trip)

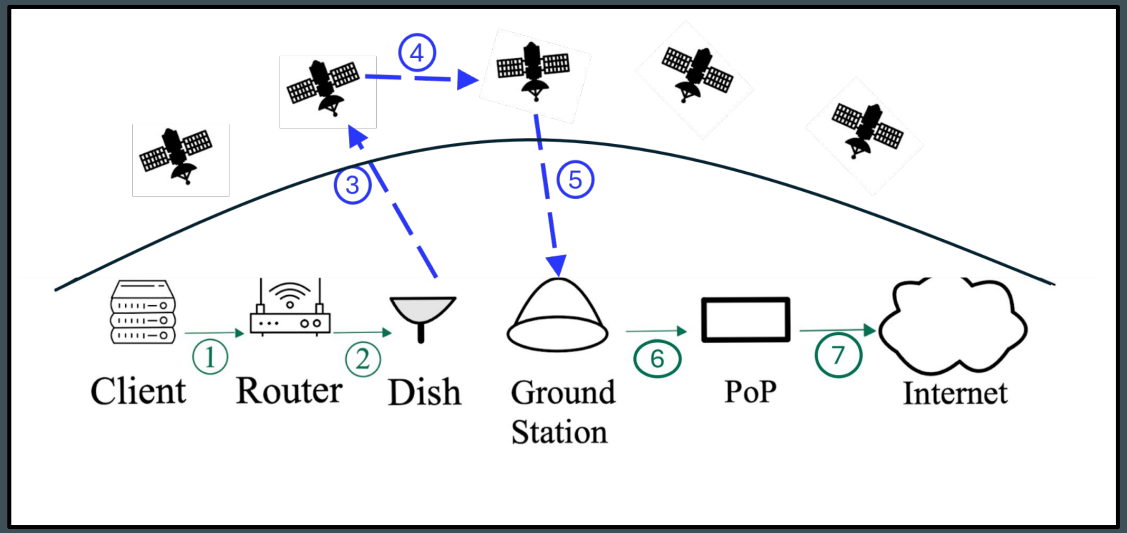
Satellite Networking Architecture

- LEO Satellites (500–2,000 km)
  - Low-latency, high-speed internet
- Ground Stations
  - Link satellites to terrestrial internet infrastructure
- User Terminals (Dish)
  - To connect users with satellites



Satellite Networking Architecture

- Inter-satellite laser links help cover remote areas and increase the overall capacity
  - Increase reliability as the network can dynamically adapt to satellite or ground station failures



Ground Stations



Image Sources: SpaceX - https://www.spacex.com/

Satellite Networking Architecture



- Small (<1 meter) and cheap
- Transmit: Kb-Band
- Receive: Ka-Band

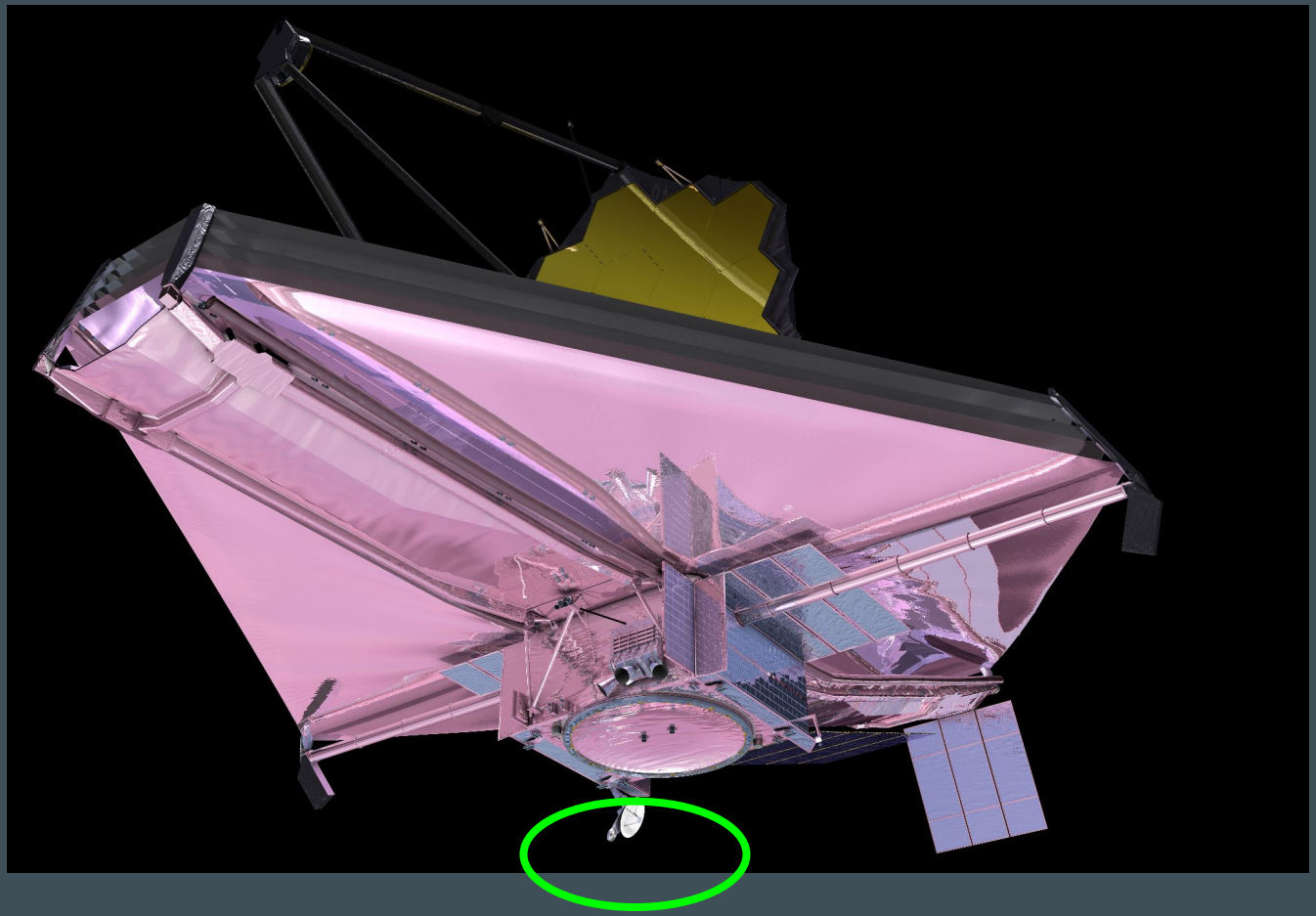


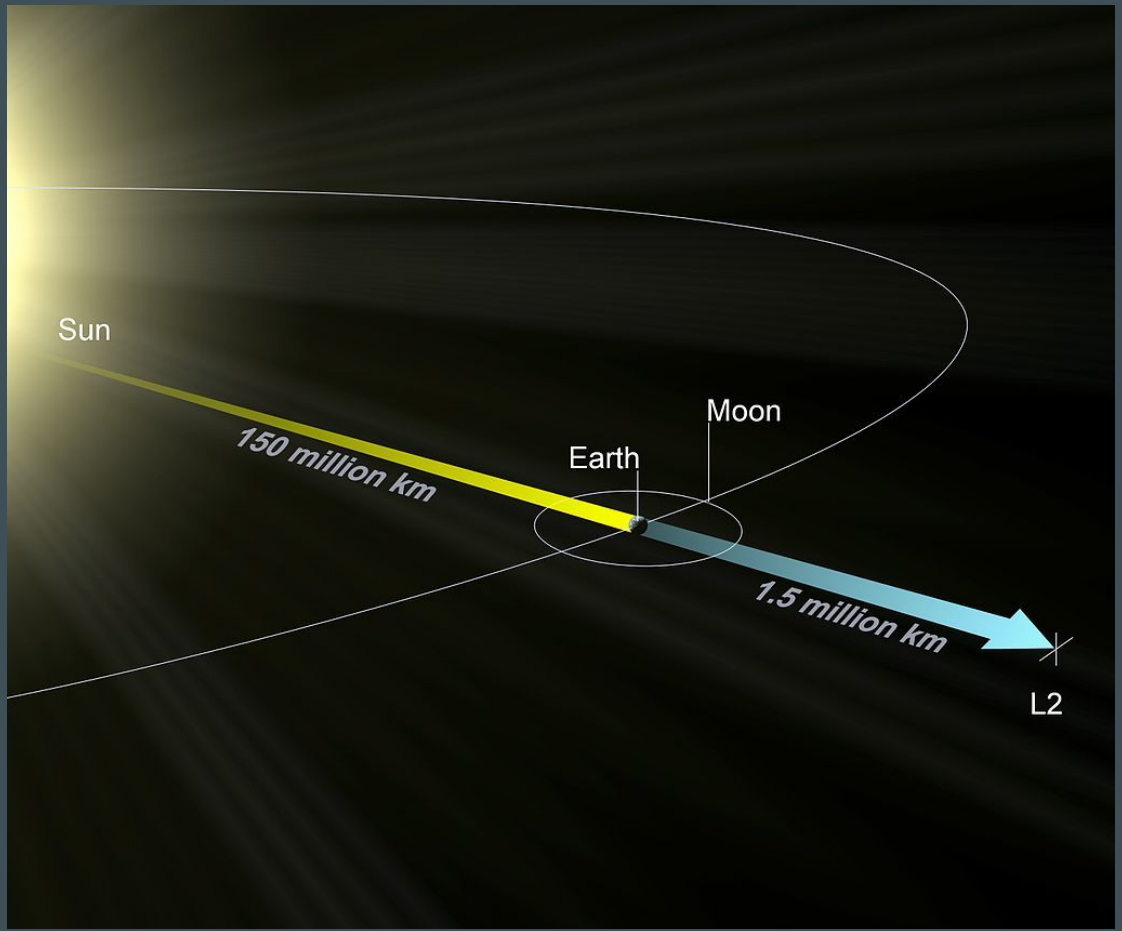
- Huge (34 metres) and expensive
- Transmit: X-Band
- Receive: X-Band and Ka-Band

NASA Deep Space Station 35, Australia



Image Sources: Canberra Deep Space Communication Complex - https://www.csiro.au/




Sun
150 million km
Earth
Moon
1.5 million km
L2

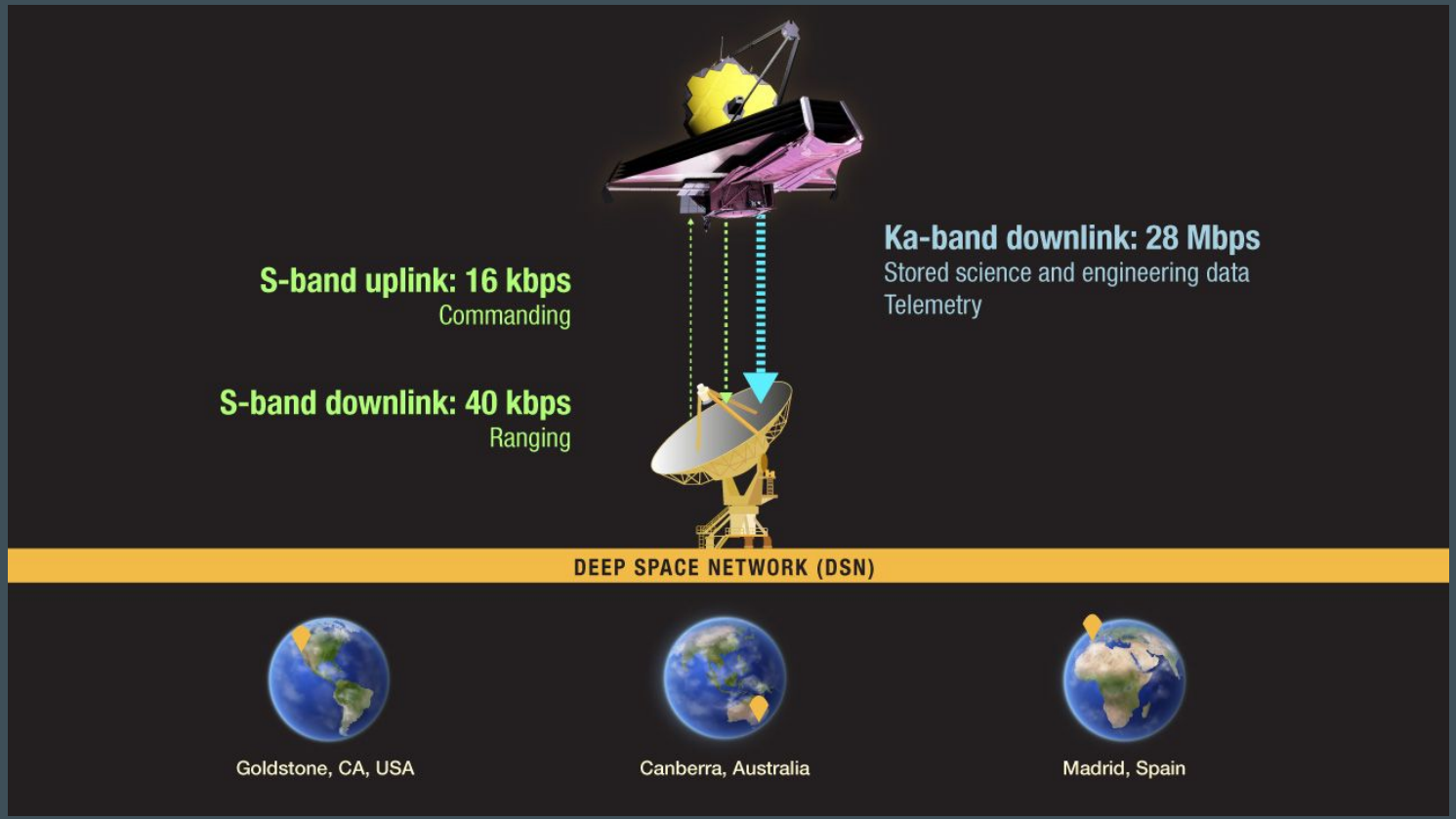


Image Sources: NASA and the Space Telescope Science Institute (STScI) - https://webbtelescope.org/contents/media/images/01HES3BMEVP7EJKJD12XE590VQ?imgsDbrtUUID=91dfa083-c258-4f9f-bef1-8f40c26f4c97&page=8

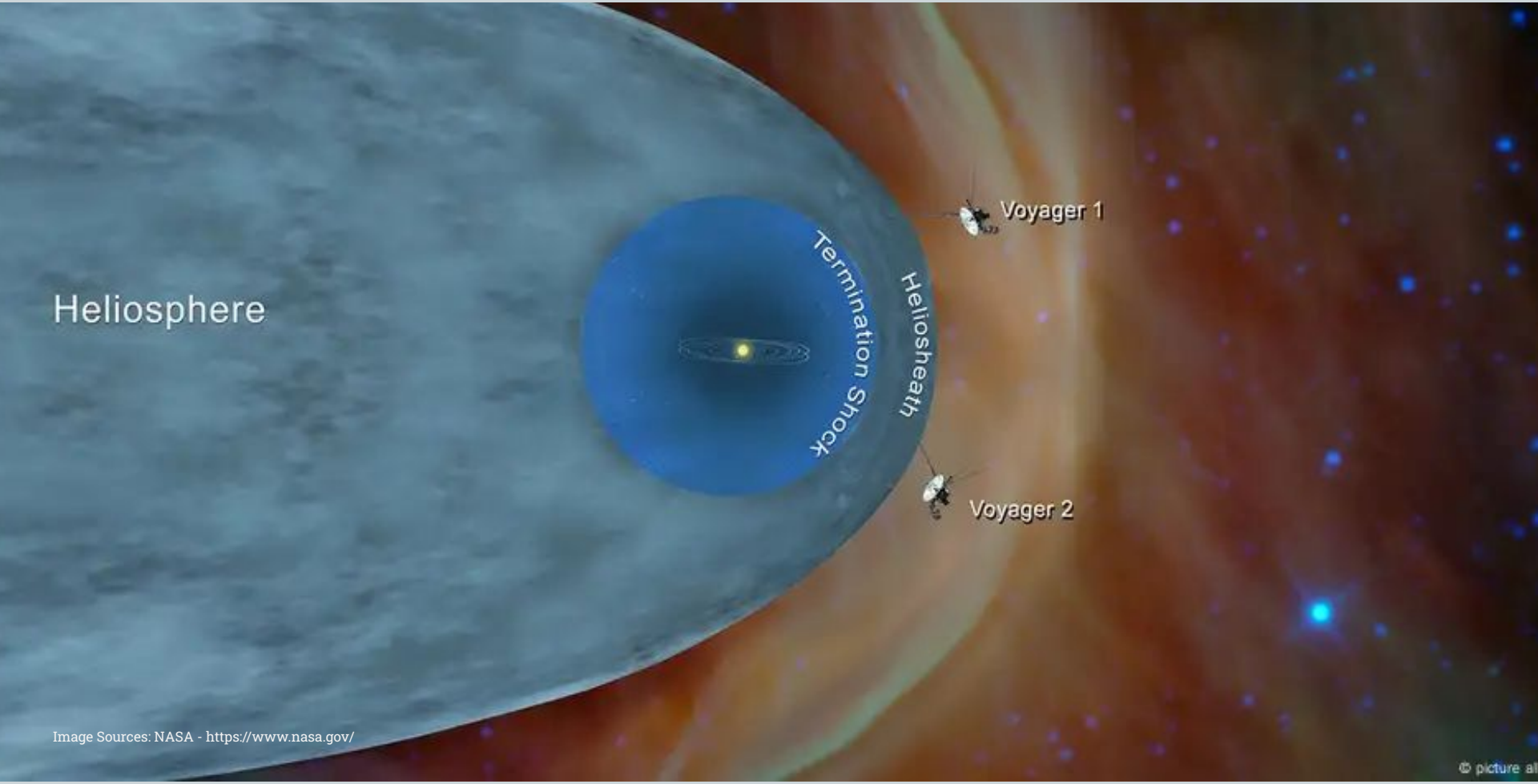


Image Sources: NASA - https://www.nasa.gov/

Voyager and Deep Space Network

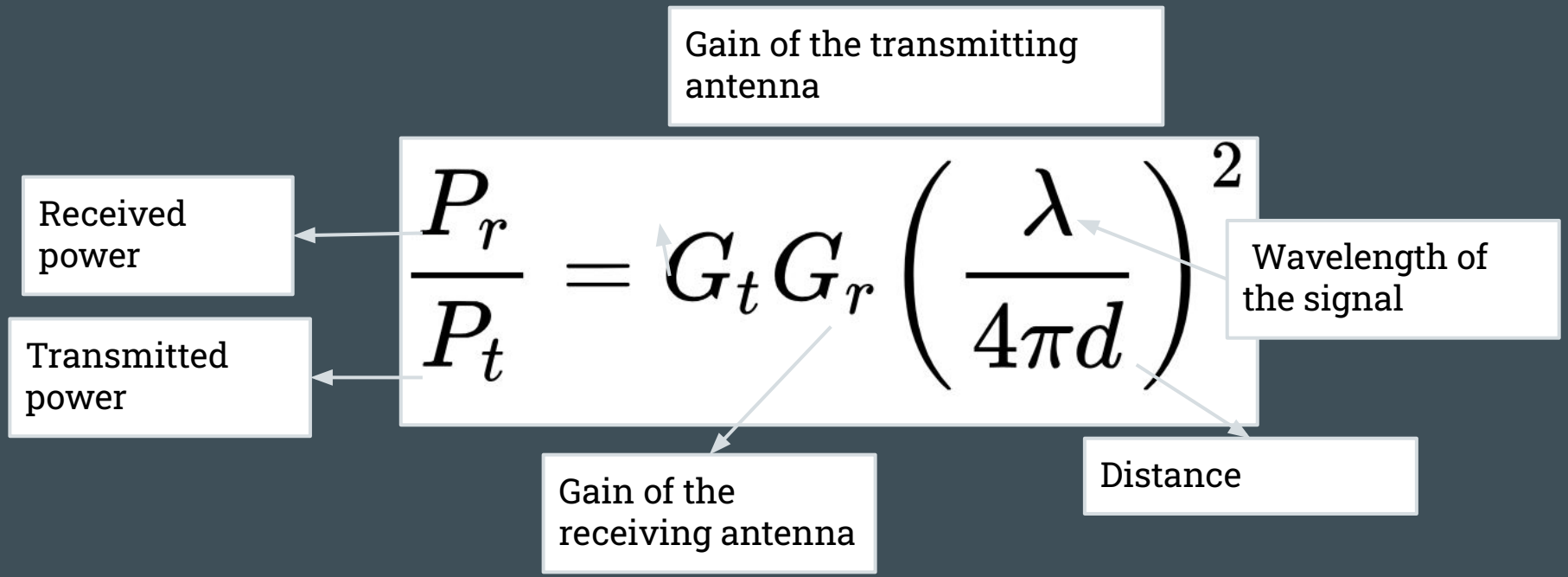


- The farthest human made object at ~24e9 km from Earth, >160x Earth-Sun distance
  - >23 hours for one-way communication, 40b/s, 34m HEF antenna

Image Sources: NASA - https://www.nasa.gov/

Voyager and Deep Space Network

- How can we even still receive a signal?



Voyager and Deep Space Network

- Using the current distance between Earth and Voyager and the available info about Voyager antennas we get:

$$P_r = \frac{25 \times 5011.87 \times 10^7 \times (0.0357)^2}{(4\pi \times 2.37 \times 10^{13})^2}$$

- $Pr=1.80\times10^{-20}$ W, barely more than a zepto watt (zW).
- This is why large antennas are needed to receive and process the incoming signal

Starlink

- On the other hand, for Starlink, the receiving power is likely in the order of several nanowatt (nW)
- Exact numbers are difficult to calculate given the proprietary nature of Starlink

Starlink

- Starlink, owned by SpaceX, has by far the biggest number of satellites in orbit (over 7000)
  - More than all other satellites combined
- Its satellites are in LEO, around 500-550 km
- Starlink satellites have a lifespan of only 5 years due to the atmospheric drag

Starlink Components

Starlink Satellites



Starlink Ground Stations



Starlink User Terminal



Image Sources: SpaceX - https://www.spacex.com/

Starlink Coverage

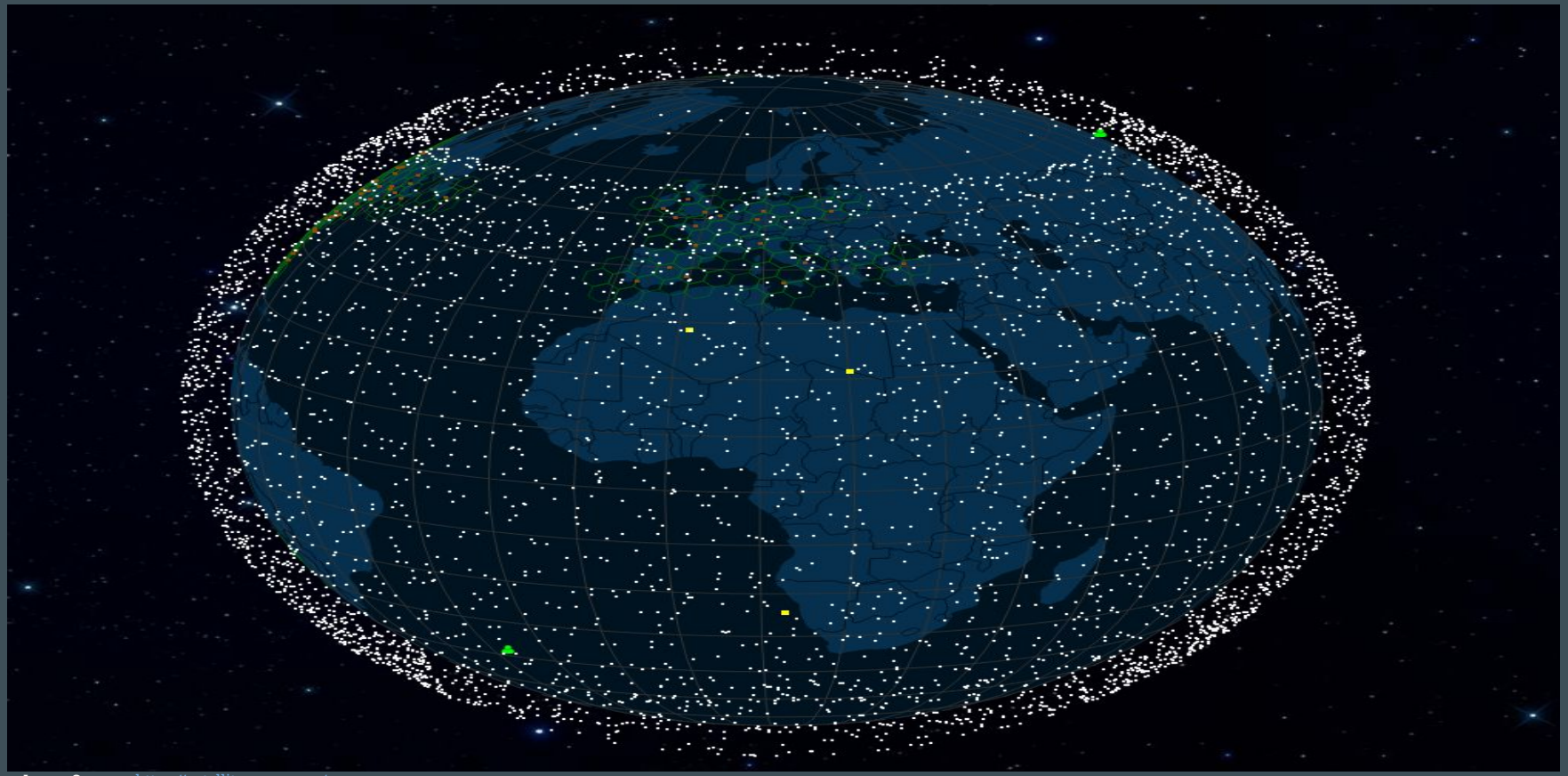
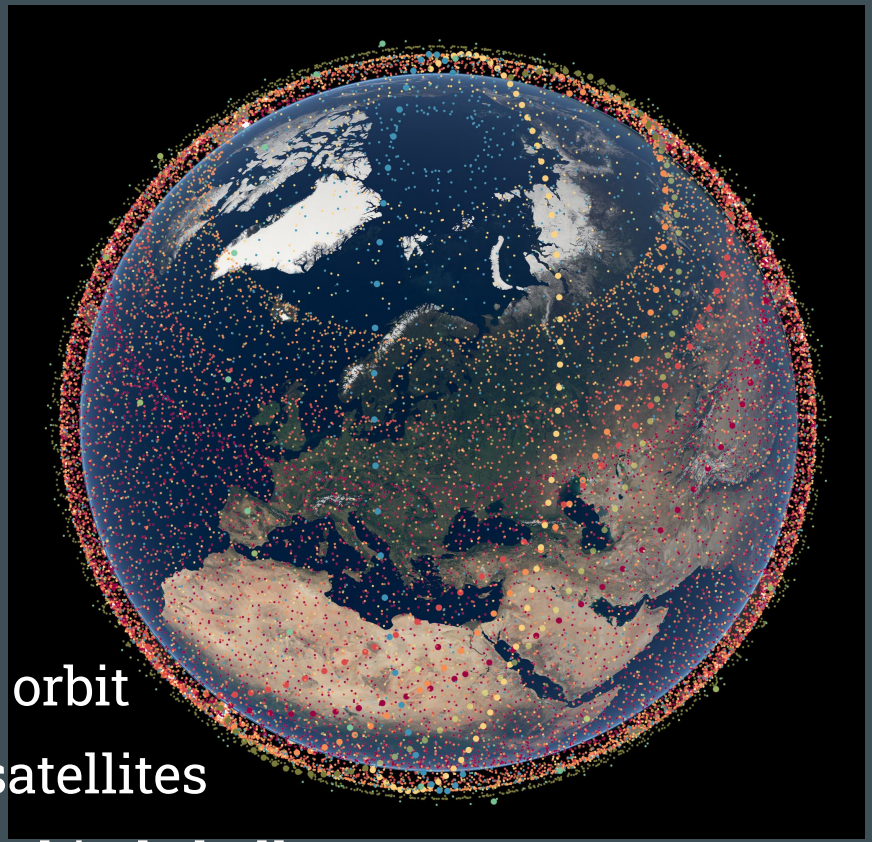
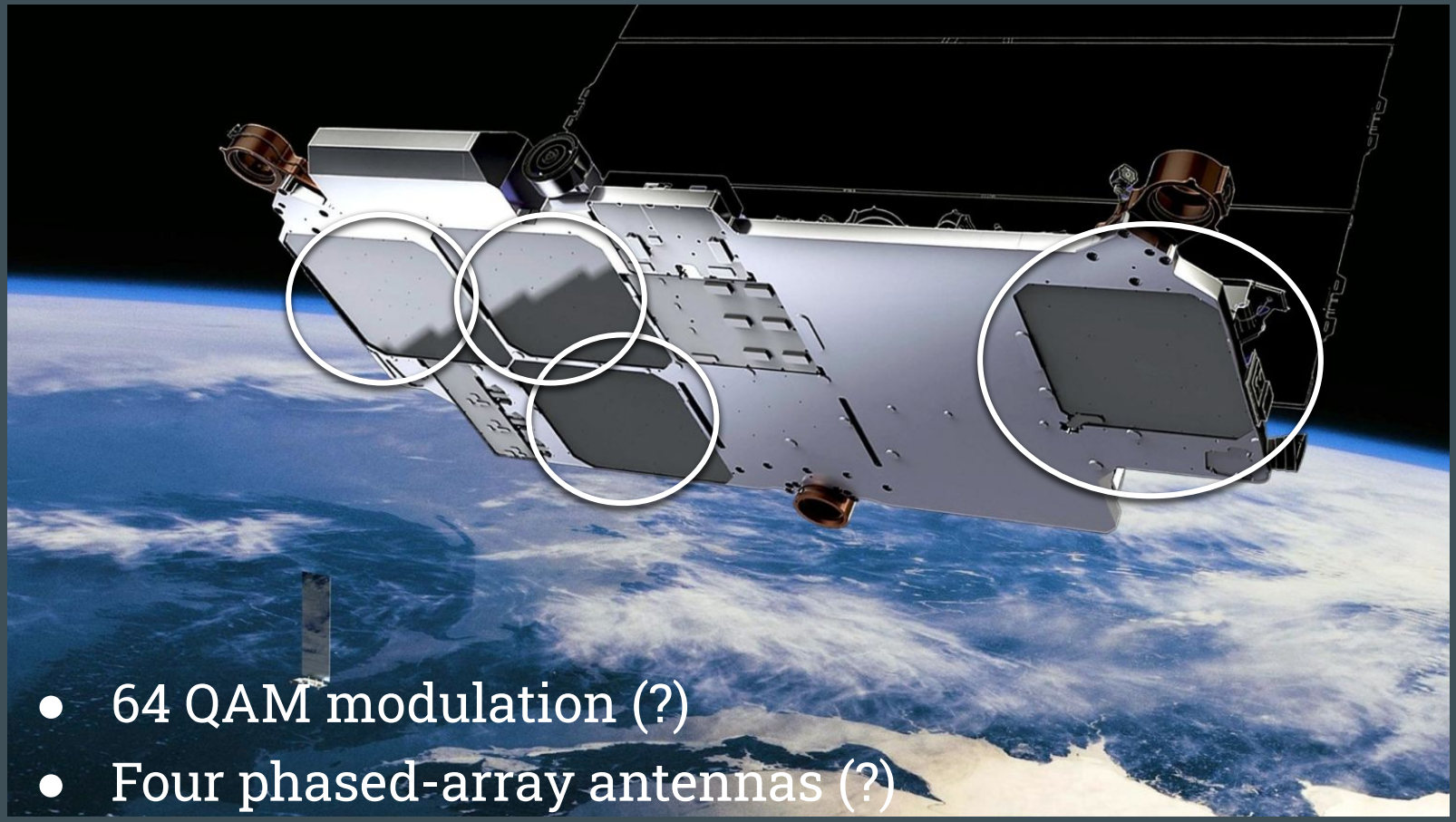


Image Sources: https://satellitemap.space/



- ~ 550 km orbit
- ~ 12’000 satellites
- multiple orbital shells

ISS: ~420 km orbit



- 64 QAM modulation (?)
- Four phased-array antennas (?)

Starlink Tech

- Each satellite (v2) has a capacity of around 80 Gb/s.
- Combining the v1 and v2 satellites, we get roughly a total capacity of 250 Tb/s
- Inter-Satellite laser communication reduces reliance on ground stations and enables fast routing
- Phased array antennas are used for beamforming and efficient communication with satellites

Starlink Direct To Cell

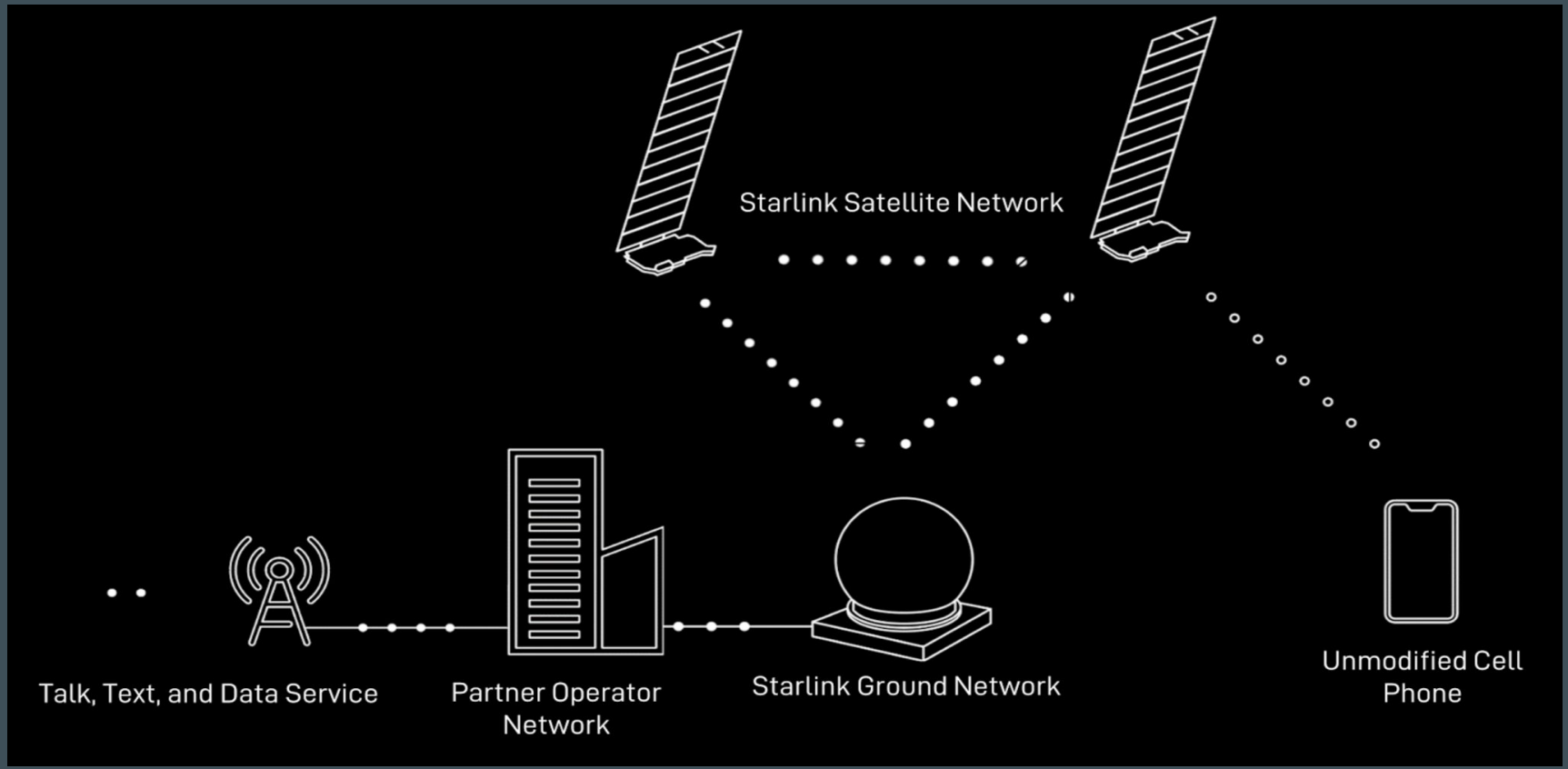


Image Sources: SpaceX - https://www.spacex.com/

Exercise: Global Navigation Satellite System

- Open in your course folder: 02_Notebooks/03_Mobile_Systems/ 03.3_GNSS.ipynb
- Run the cells and answer the questions.
- Save the completed notebook and upload it to Moodle.

